Weekly Team Check-In 3: Evaluation & Testing

KPMG × Columbia QMSS Practicum (5052) · Since Check-In 2: Nov 2 – Nov 8, 2026 (KPMG meeting 11/5; next meeting 11/12)

Workstream	Evaluation & Testing: measure whether the SEC-filings assistant's answers can be trusted
Team leads	Chuan-Yeu (Thomas) Song, Binyu Wang
Members and roles	Shenyi Feng (numeric accuracy) · Fei Huang (citation support) · Shuhan Bian (retrieval metrics; RAG liaison) · Yupin Wang (refusal/abstention) · Ruofei Duan (GraphRAG metrics; GraphRAG liaison) · Bo Yuan (latency, cost, consistency) · Binyu Wang (reasoning steps; Agents liaison) · Thomas Song (gold set; Data and KPMG liaison)
Where the work is	GitHub thomassong0604-oss/kpmg-eval → Week 7/ (presented 11/5) and Week 8/ (for 11/12); 82 automated tests passing; FOR_TEAMS_W8.md has one section per team

1. Progress since the previous check-in

Moved the gold set onto the Data team's ten companies and database (Gold Set v2)

- **Matched our filing list to Data's database** (21 filings as of 11/03). Our earlier list matched 20/21; the difference was Micron's FY2026 10-K (filed 10/09), which we had as FY2025. Gold Set v2 matches **21/21**, and all 21 are in Data's 233-filing 2021–2026 inventory.
- **Gold Set v2: 155 questions, 143 answerable**, replacing the 139-question round-1 draft. Every answerable question is on the ten companies (Microsoft, the MVP, has 92; the others 9–16 each). Each question now records the route it needs: **104 finance data, 31 text, 11 relationship, 8 no tool, 1 both**.
- **7 relationship questions from Exhibit 21** (subsidiary lists: Apple 19, Casey's 9, Micron 21, each transcribed and read twice). With the 4 drafted with GraphRAG, **11 questions now require GraphRAG**; 4 of them test invented or false links.
- **Checks:** 55/55 re-reads agree; 18/18 quarterly figures sum to the reported totals; 160/160 grader labels right on the new numeric questions; all nine other companies' income statements, read from the filings, match the gold.
- **SEC data issues turned into test questions:** NetApp tags 3 of its 10-Qs one fiscal year behind its own text; McKesson's and Medtronic's statement net income includes noncontrolling interests (McKesson 5,099 vs 4,762), so questions now say which one and the other is scored as a wrong answer.

Measured the Agents' decisions and routing (KPMG 10/29 request)

- **Answer / clarify / abstain / block:** drafted the rules (POLICY_answer_clarify_abstain.md; no tool calls on off-topic questions, as KPMG asked) and a 24-item test suite. On 179 questions, answering everything gives **100% unsafe answers** on questions that should not be answered and 8 off-topic tool calls; a simple scope check gives **96.1% right decisions**, 0 off-topic calls, 0 needless refusals, 13.9% unsafe (5 questions left).
- **Routing:** scorer for any router. Text search only: 20% of questions get the tool they need; our keyword router: 84% at 1.05 calls/question; calling every tool: 95% at 3.00 calls (2.05 unneeded).

RAG and GraphRAG side by side; real-user wording (KPMG 11/5 requests)

- **Same filing, each team's own code, 84 Microsoft 10-Q questions.** Needed figures in the top 8: **RAG 49/50, GraphRAG 25/50**; prose passages: **18/27 vs 0/27** (GraphRAG's structured search stores numbers only). GraphRAG shows the wrong-period figure less often (**19/36 vs 33/36**) because its facts carry their period.
- **Where each does better:** relationship questions favour GraphRAG. RAG does not index Exhibit 21, while traversing GraphRAG's own Exhibit 21 rows answers all 9 exactly (their rows match our gold reads 62/62). RAG is better at prose and at finding numbers. We traced GraphRAG's 25 misses so it can fix them: 13 ranked just below the top 8 (total revenue ranks 10th behind "Microsoft 365" product rows), 10 calculations needing several facts from one call, 2 wording.
- **Typos and messy wording:** 427 rewrites with the same meaning. With no company selected, the scope check drops from **98% to 19%** on misspelled names; matching tickers and near-miss spellings restores **94%**. Misspelled finance words ("revnue") cost both searches 9 of 70 questions.
- **Routing for the demo:** chart axes labelled, keyword router and "unneeded call" defined (we corrected the minutes' wording), 7 worked examples, and every rate labelled as based on the current 155 questions.

Status of Check-In 2's next steps

Planned	Status	Detail
1. Score numeric accuracy with RAG's fixed reader, by route	Partly	Reader built and tested with a stub (gpt-4o-mini, temperature 0, RAG's prompt); every scorecard reports accuracy by route. **Not yet run: no API key** (see blockers).

Planned	Status	Detail
2. Complete round-1 gold set and anchors	Re-scoped	Replaced by Gold Set v2. RAG's draft anchors reviewed (results/rag_anchor_fixes.csv: 10 drop incl. 8 Amazon/Meta, 2 remove, 25 narrow, 17 with ready replacements); Exhibit 21 anchors in. Text anchors for new lookups wait on Data's chunks.
3. GraphRAG and Agents hand-offs	Partly	Hops sheet extended to 165 questions; GraphRAG's real fact loader and search now run in our harness (no longer only the mock), and their Exhibit 21 output checks out against our reads (62/62); graph arm still needs a database. Decision rules sent to Agents with a request for a decision field in the AnswerBlock.

Feedback received and how we responded

Feedback	Response
KPMG 10/29: match the filing list with Data; bring in Exhibit 21	21/21 match; Micron corrected; 7 Exhibit 21 questions
KPMG 10/29: when should the Agent answer, ask, or abstain; compare routing with combined retrieval	Rules, 24-item suite and two scorers delivered; results above
KPMG 11/5: run RAG and GraphRAG side by side; show where each is better	Done on the full Microsoft 10-Q set plus 9 relationship questions
KPMG 11/5: label routing axes; define keyword router and unneeded calls; demo examples; say results use 155 questions	Done (Week 8 slides 4–5, ROUTING_EXAMPLES.md)
KPMG 11/5: test typos and messy wording	427 rewrites through scope check, router and both searches
RAG: what to do with 8 Amazon/Meta questions; freeze a data snapshot	Drop them (outside the ten); proposed Data's 11/03 database as the round-1 snapshot

Decisions and deliverables

- **Decisions:** Gold Set v2 is the round-1 set (Gold Set v1 stays frozen for comparability); comparison questions name the 11/03 snapshot date so later loads cannot change answers; results are reported at evidence level, labelled as such, until a reader can run; every result pins the commit of each team's code it used.
- **Delivered:** 6 Eval slides for 11/5, 7 for 11/12 (with speaker notes); Week 7 and Week 8 code, data and results on GitHub; FOR_TEAMS notes per week; Slack updates to RAG (anchor review, snapshot) and GraphRAG (how their retrieval reads the database, where our files are).

2. Current priorities

- **Round 1 on the frozen snapshot (week 8):** score RAG and GraphRAG on Gold Set v2 by route and question type, with cost and latency. Shenyi Feng, Shuhan Bian, Ruofei Duan, Bo Yuan.
- **11/12 KPMG meeting:** present the side-by-side and walk through the routing examples. Thomas Song, Binyu Wang.
- **Gold set coverage:** text anchors for new questions; more questions where the set is thin; the messy rewrites as a robustness slice. Thomas Song, Fei Huang, Yupin Wang.

3. Next steps before the next check-in

#	Action	Owner(s)	Done when
1	Run round 1 on the 21-filing snapshot: RAG (retriever + fixed reader if a key is available) and GraphRAG's structured search on Gold Set v2, by route and question type; log latency and cost per question	Shenyi Feng (lead), Shuhan Bian (RAG), Ruofei Duan (GraphRAG), Bo Yuan (cost, latency)	Round-1 scorecard on GitHub (Week 9/) and shared in the RAG and GraphRAG threads; answer-level if a key exists, otherwise evidence-level, labelled
2	Finish anchors and widen thin areas: text anchors once Data's chunks land; more questions on non-Microsoft companies and on the "both" (1) and "no tool" (8) routes; add the 427 rewrites as a robustness slice	Thomas Song (lead), Fei Huang (anchors), Yupin Wang (unanswerable)	Every answerable question has a verified anchor; Gold Set v2.1 published with a hash and change log

#	Action	Owner(s)	Done when
3	Score the Agents' real orchestrator instead of our stand-in rules: routing and decision scorers, agreed expected routes, ticker and near-miss name matching	Binyu Wang (lead), Yupin Wang	Orchestrator scores next to our reference rules on the same 179 questions

4. Challenges and blockers

- **No API key yet.** The fixed reader is ready but cannot run, so all results are at evidence level (is the right number or passage in what the system retrieved?). **Risk:** round 1 cannot report answer accuracy, KPMG's headline metric. **Mitigation:** evidence-level results are exact and reproducible; key requested (section 5).
- **GraphRAG's graph cannot be scored yet.** It needs a Neo4j database we cannot reach. Their new Exhibit 21 output for the ten companies (pushed 11/8 on a branch) agrees with our gold reads on all 62 rows we hold, but is not yet merged or loaded into the graph. **Mitigation:** we scored the structured arm with their own code (latest main), traverse their Exhibit 21 rows ourselves, and pin the exact commits.
- **Thin, skewed gold set.** Microsoft has 92 of 143 answerable questions; only 1 question needs both tools and 8 need none, so per-route rates on these are examples, not estimates. KPMG asked us to say this, and every number is labelled. Next step 2 addresses it.
- **Side-by-side limited to one filing.** RAG built its round-1 corpus for all 21 filings on 11/5 (its freeze list matches ours 21/21), but the chunks live in Data's database, not the repo. Getting them unblocks both text anchors and a ten-company comparison.
- **SEC data pitfalls, including one we got wrong.** The SEC data API lagged Micron's FY2026 10-K, so on 10/29 we said it was not in the corpus; we corrected this on 11/5 and now check the filing documents, not the API alone. NetApp's mislabelled fiscal year can mislead every team's filters.
- **Real-user wording.** Misspelled company names and finance words break simple matching (above). The fixes are small but sit with Agents, RAG and GraphRAG before the demo.
- **Coordination.** Our expected routes and decision rules are Eval's labels until Agents review them. The 11/5 minutes described "unneeded calls" differently from our metric; we clarified it in writing so no team optimises the wrong thing.

5. Cross-team coordination and support needed

From	Request
Data	Keep the 11/03 freeze list (21 filings, matches ours 21/21) fixed through round 1, with no reloads. Thank you for the 233-filing inventory, which made the filing match possible.
RAG	Share the round-1 chunks for the 21 filings (privately; Data keeps exports off GitHub); let the router send Exhibit 21 questions to GraphRAG; an API key for the fixed reader (or who holds one). Thank you: your retriever ran in our harness unchanged.
GraphRAG	A Neo4j endpoint so we can score the graph arm; merge and load the new Exhibit 21 output; treat "three/six months" as period words (they match Microsoft 365 rows and push totals out of the top 8); two concept aliases. Thank you: your fact loader ran unchanged and your Exhibit 21 rows matched ours.
Agents	Run your orchestrator through our routing and decision scorers; review our expected routes; ticker and near-miss name matching in the scope check; a decision field in the AnswerBlock.
KPMG	Confirm the unneeded-call definition and demo examples show what you want; examples of real analyst wording or common misspellings, to make the messy-wording set realistic; a target accuracy for round 1, if you have one.
Instructor / TAs	An API budget or key for the fixed reader; whether round 1 may report evidence-level scores for components without a reader.